

CS772 Assignment 2

Transliteration

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Transliteration ?

Transliteration is the process of converting text from one script to another while **preserving pronunciation** rather than meaning.

It helps readers of one script accurately pronounce words written in another.

Hindi Transliteration (Roman → Devanagari)

Converting Hindi words written in Roman script (English letters) into Devanagari script (हिन्दी).

Why Transliteration is Used:

- Ease of communication: People often type Hindi using Roman letters on phones/computers.
- Improves readability: Converts informal Romanized Hindi into proper script.
- Useful for NLP tasks: Enables search, translation, and speech systems to understand Hindi text better.
- Preserves language integrity: Keeps pronunciation and script aligned.

Data Downloading and Cleaning

100k samples used for training out of 1.3 million

Sampling Method : Weighted sampling using a mix of:

- Score rank (70%) — higher quality pairs prioritized
- Word length rank (30%) — ensure diversity in word size

Additional stratification:

- Binned by word length (short, medium, long, xlong)
- Binned by score (very_low → very_high)
- Considered consonant-vowel pattern diversity

Source - Aksharantar hindi corpus

Cleaning Steps

Step	Purpose
Keep only alphabetic words	Remove punctuation/numbers
Drop duplicates	Avoid repetition
Remove low-quality pairs (bottom 5%)	Ensure transliteration reliability
Filter word length (3–20 chars)	Remove noise/typos

LSTM Based transliteration

Read about sequence-to-sequence (Seq2Seq) and encoder–decoder LSTM models for transliteration. Key concept: model learns character-level mappings between Roman and Devanagari scripts.

Method	Accuracy	CER	Comments
Greedy Decoding	0.446	0.165	Fast but prone to local optima
Beam Search (w=5)	0.466	0.157	Slightly better accuracy, smoother outputs

Model Performance

- Overall Accuracy: **46.6%**
- Character Accuracy: **75.9%**
- Avg. Similarity: **0.86**
- Throughput: **6 words/sec**

Trend:

Longer words → slightly higher accuracy (due to richer context).

Model & Training Setup

- Architecture: **Bi-LSTM Encoder → Attention Decoder**
- Input: Roman characters → Output: Devanagari sequence
- Trained on **100K cleaned samples** from Aksharantar (Hindi)
- Optimized using **Cross-Entropy Loss**
- Decoding: **Greedy** vs **Beam Search (width=5)**

Error Patterns

Confusion between similar vowels (ा ↔ ी, ू ↔ ु)

Missing or extra matras (diacritics)

English abbreviations mis-transliterated (“win” → “विन” instead of “डब्ल्यूआईएन”)

LSTM Analysis

OVERALL METRICS

```
total_samples      : 10112
accuracy           : 0.4659
average_cer        : 0.1574
average_similarity  : 0.8582
average_char_accuracy : 0.7587
average_edit_distance : 0.9590
median_edit_distance : 1.0000
average_inference_time_ms : 165.5472
median_inference_time_ms : 179.2254
throughput_words_per_sec : 6.0406
```

PERFORMANCE BY LENGTH

length_range	count	accuracy	avg_cer	avg_similarity	avg_edit_distance
1-3	447	0.384787	0.416480	0.709242	1.165548
4-6	4989	0.473442	0.171945	0.844829	0.871317
7-10	3677	0.460974	0.121675	0.882704	0.973348
11-15	919	0.482046	0.101636	0.899881	1.238303
16-100	80	0.487500	0.080991	0.919685	1.400000

ERROR PATTERNS

```
length_mismatch      : 916
too_short            : 307
too_long             : 609
off_by_one           : 2829
off_by_two           : 1548
```

TOP CHARACTER CONFUSIONS

```
ा → र: 228 times
ी → ः: 208 times
र → l: 203 times
ट → त: 195 times
ा → न: 170 times
् → र: 170 times
ं → न: 155 times
ड → द: 151 times
र → ्र: 148 times
ॠ → ्रु: 146 times
```

TOP 10 WORST ERRORS

```
Source:      qiwi
Reference:    क्यूआईडब्ल्यूआई
Prediction:    क्रिवी
Edit Dist:    14

Source:      phwcs
Reference:    पीएचडब्ल्यूसीएस
Prediction:    फफव्स
Edit Dist:    13

Source:      qwvga
Reference:    क्यूडब्ल्यूवीजीए
Prediction:    क्वग्गा
Edit Dist:    12

Source:      win
Reference:    डब्ल्यूआईएन
Prediction:    विन
Edit Dist:    10

Source:      dow
Reference:    डीओडब्ल्यू
Prediction:    डोव
Edit Dist:    9
```

Transformer based transliteration

Model Background : Transformer-based Seq2Seq model

- Unlike LSTMs, which process tokens sequentially, Transformers: Look at all positions at once (parallel attention)
- Capture long-range dependencies better
- Are generally faster to train but heavier at inference

Reading : “Attention Is All You Need” (Vaswani et al., 2017)

Metric	Transformer	LSTM
Accuracy	0.478	0.466
Char Accuracy	0.758	0.759
CER	0.161	0.157
Edit Distance	0.956	0.959
Throughput (words/sec)	5.41	6.04

Method	Accuracy	CER ↓	Notes
Greedy Decoding	0.462	0.169	Fast but misses alternate spellings
Beam Search (width = 5)	0.478	0.161	Keeps multiple candidates — better transliterations

Interpretation:

- Transformer slightly **outperforms LSTM** on word-level accuracy (+1.2%).
- Character-level accuracy is roughly **the same** (0.758 vs 0.759).
- Slightly **slower inference** (184 ms per word vs 165 ms).
- Overall, **Transformer is more stable and context-aware**, but not drastically better on small datasets.

Transformer Analysis

OVERALL METRICS

```
-----
total_samples           : 10112
accuracy                : 0.4778
average_cer             : 0.1608
average_similarity      : 0.8581
average_char_accuracy   : 0.7580
average_edit_distance   : 0.9565
median_edit_distance    : 1.0000
average_inference_time_ms : 184.8126
median_inference_time_ms : 170.5719
throughput_words_per_sec : 5.4109
```

PERFORMANCE BY LENGTH

```
=====
length_range  count  accuracy  avg_cer  avg_similarity  avg_edit_distance
-----
1-3           447   0.340045  0.542133      0.651979      1.514541
4-6           4989  0.484867  0.171868      0.845572      0.870114
7-10          3677  0.481099  0.116615      0.887712      0.932554
11-15         919   0.497280  0.096629      0.904701      1.173014
16-100        80    0.437500  0.103317      0.896683      1.837500
```

ERROR PATTERNS

```
=====
length_mismatch       : 950
too_short             : 312
too_long              : 638
off_by_one            : 2772
off_by_two            : 1439
```

TOP CHARACTER CONFUSIONS

```
=====
ा → र: 226 times
ी → ि: 222 times
र → I: 190 times
ट → त: 182 times
ड → द: 173 times
ा → न: 158 times
् → र: 158 times
्र → ः: 148 times
न → : 143 times
ं → न: 137 times
```

TOP 10 WORST ERRORS

```
=====
Source:      qiwi
Reference:   क्यूआईडब्ल्यूआई
Prediction:   क्रिवी
Edit Dist:   14
-----
Source:      qwvga
Reference:   क्यूडब्ल्यूवीजीए
Prediction:   कखा
Edit Dist:   13
-----
Source:      phwcs
Reference:   पीएचडब्ल्यूसीएस
Prediction:   फक्स
Edit Dist:   12
-----
Source:      win
Reference:   डब्ल्यूआईएन
Prediction:   विन
Edit Dist:   10
-----
Source:      pwid
Reference:   पीडब्ल्यूआईडी
Prediction:   प्विड
Edit Dist:   10
```

LLM transliteration

Models Evaluated:

- **NVIDIA Mistral NeMo (12B)** – smaller, faster model for baseline transliteration.
- **NVIDIA Mistral Small (24B)** – larger model with improved contextual learning and richer vocabulary.

Temp	Top -p	(12B)	(24B)	Observation
0.3	0.8	0.31	0.43	Slightly rigid outputs
0.7	0.9	0.33	0.44	Balanced phonetic precision and variation
1.0	1.0	0.29	0.40	More creative but often phonetic drift

Metric	Mistral NeMo (12B)	Mistral Small (24B)
Accuracy	32.8%	44.3%
Character Accuracy	66.6%	74.4%
Character Error Rate (CER)	0.2169	0.1827
Similarity Score	0.7968	0.8467
Edit Distance ↓	1.41	1.13
Inference Time (ms)	361	416

LLM Nemo 12B analysis

OVERALL METRICS

total_samples	: 10112
failed_requests	: 0
accuracy	: 0.3278
average_cer	: 0.2169
average_similarity	: 0.7968
average_char_accuracy	: 0.6660
average_edit_distance	: 1.4142
median_edit_distance	: 1.0000
average_inference_time_ms	: 361.3027
median_inference_time_ms	: 355.2123
throughput_words_per_sec	: 2.7678

ERROR PATTERNS

length_mismatch	: 1302
too_short	: 708
too_long	: 594
off_by_one	: 2845
off_by_two	: 2066
empty_predictions	: 0

TOP 10 WORST ERRORS

Source: distributary
Reference: डिस्ट्रीब्यूटरी
Prediction: distri'bju:təri
Edit Dist: 18

Source: dublin
Reference: डबलिन
Prediction: डब्ल्यूब्ल्यूब्ल्यू
Edit Dist: 16

Source: distributary
Reference: डिस्ट्रीब्यूटरी
Prediction: distri'bju:təri
Edit Dist: 15

Source: mpwppcl
Reference: एमपीडब्ल्यूपीपीसीएल
Prediction: म्पव्चल
Edit Dist: 14

Source: communications
Reference: कम्यूनिकेशन्स
Prediction: sampark
Edit Dist: 13

TOP CHARACTER CONFUSIONS

ं → न: 402 times
ी → ः: 397 times
ा → र: 321 times
ड → द: 251 times
र → : 245 times
ि → ी: 220 times
र → I: 211 times
न → I: 204 times
ा → न: 202 times
ॠ → उ: 174 times

PERFORMANCE BY LENGTH

length_range	count	accuracy	avg_cer	avg_similarity	avg_edit_distance
1-3	447	0.409396	0.318046	0.729701	0.914989
4-6	4989	0.365805	0.230694	0.786488	1.180196
7-10	3677	0.287463	0.199437	0.808069	1.605929
11-15	919	0.245919	0.166176	0.837339	2.020675
16-100	80	0.300000	0.172228	0.831667	3.012500

LLM Small 24B analysis

OVERALL METRICS

total_samples	: 10112
failed_requests	: 0
accuracy	: 0.4432
average_cer	: 0.1827
average_similarity	: 0.8467
average_char_accuracy	: 0.7439
average_edit_distance	: 1.1350
median_edit_distance	: 1.0000
average_inference_time_ms	: 416.4669
median_inference_time_ms	: 386.2234
throughput_words_per_sec	: 2.4012

ERROR PATTERNS

length_mismatch	: 1000
too_short	: 361
too_long	: 639
off_by_one	: 2768
off_by_two	: 1658
empty_predictions	: 0

TOP CHARACTER CONFUSIONS

ं → न: 354 times
ी → ि: 253 times
र → I: 227 times
ा → र: 220 times
र → : 198 times
ड → द: 195 times
ि → ी: 191 times
त → ट: 181 times
न → I: 176 times
ा → न: 171 times

TOP 10 WORST ERRORS

Source:	suchnaah
Reference:	सूचना:
Prediction:	सुचनासम्पूर्णाशिक्षासम्पूर्णाशिक्षा
Edit Dist:	31
Source:	rasaayanon
Reference:	रसायनो
Prediction:	रसायनकेमिस्ट्रीफिजिक्सबायोलॉजीमैथेम
Edit Dist:	29
Source:	retreate
Reference:	रीट्रीट
Prediction:	रिट्रीटसाइंसएजुकेशनटेक्नोलॉजीइंटरने
Edit Dist:	29
Source:	nations
Reference:	नेशंस
Prediction:	नेशनयूनिटीडाइवर्सिटीडेमोक्रेसीफ़
Edit Dist:	28
Source:	najaraandaaajh
Reference:	नजरअंदाज़
Prediction:	नजारान्दाजसच्चिनमहात्मागांधीइंडिया
Edit Dist:	28

PERFORMANCE BY LENGTH

length_range	count	accuracy	avg_cer	avg_similarity	avg_edit_distance
1-3	447	0.487696	0.331469	0.772987	0.941834
4-6	4989	0.449188	0.208048	0.830064	1.058328
7-10	3677	0.427250	0.151214	0.864285	1.213761
11-15	919	0.447225	0.106855	0.896899	1.290533
16-100	80	0.512500	0.092185	0.909447	1.587500

Comparison

Model	Word Accuracy (%)	CharAcc (%)	Char F1 (%)	CER (↓)
LSTM	46.6	75.9	74.5	0.157
Transformer	47.8	75.8	74.3	0.161
NVIDIA NeMo 12B	32.8	66.6	64.8	0.217
NVIDIA Small 24B	44.3	74.4	72.9	0.183

- Transformer slightly outperforms LSTM in word-level accuracy.
- Large LLM (24B) improves phonetic accuracy over smaller 12B model, but still below classical models.
- LLMs are slower (higher inference times) but capable of context-aware predictions.

Best classical model: Transformer (~48% accuracy).

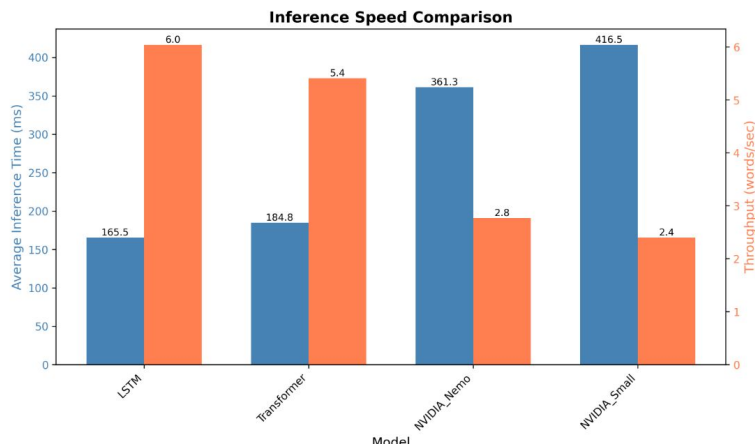
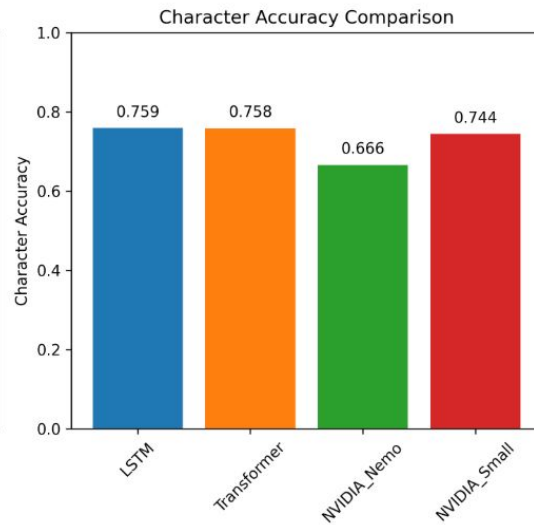
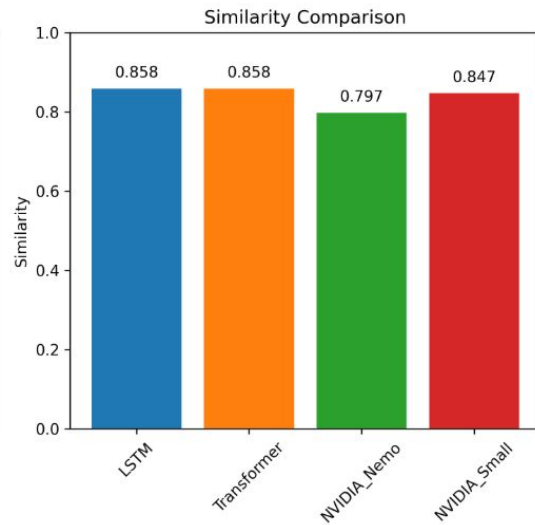
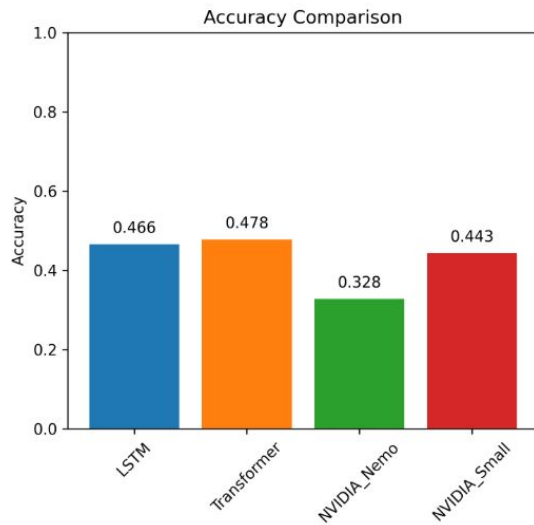
Best LLM: 24B (44% accuracy, stronger character-level consistency).

Most challenging cases:

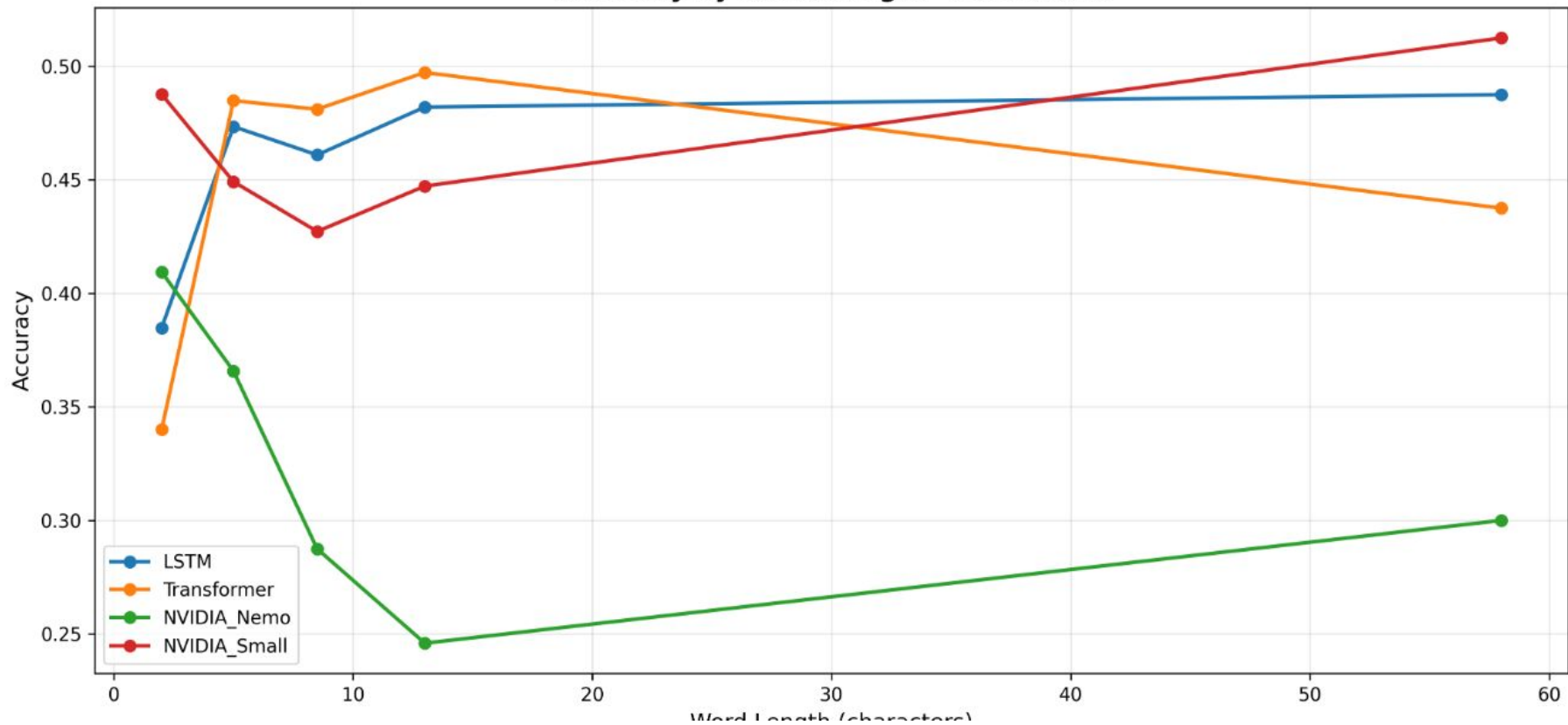
- Short acronyms
- Long compound words
- Rare consonant-vowel sequences

Difficult Characters / Sequences

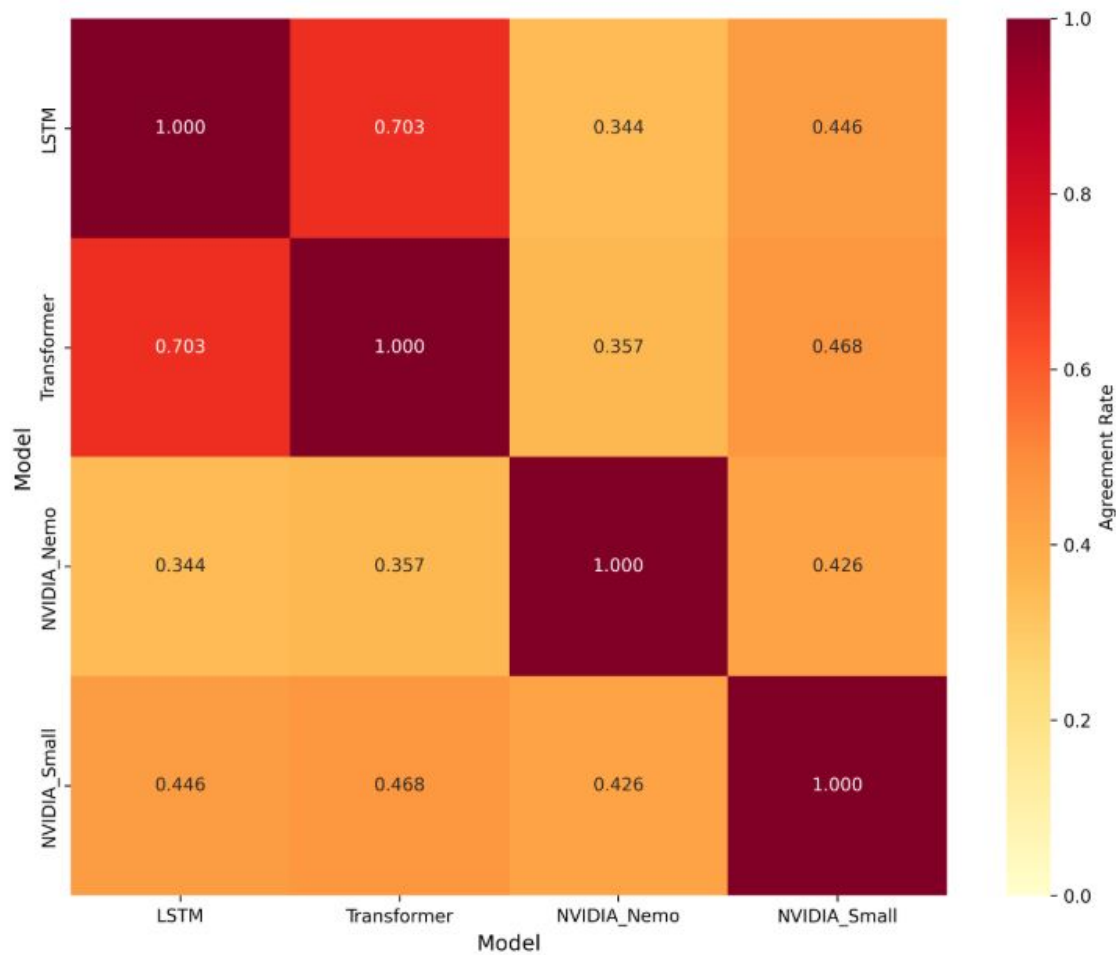
- Frequently mis-transliterated phonemes:
 - ं ↔ न, ी ↔ ि, ा ↔ र, ट ↔ त, ड ↔ द
- **Foreign/abbreviation words** (e.g., acronyms like “NCRWC”, “WIN”) are error-prone.
- **Long or compound words** increase error rate (off-by-one/two or length mismatches).
- LLMs sometimes over-generate or insert semantic words, leading to very long incorrect outputs.



Accuracy by Word Length - All Models



Model Agreement Matrix



CS772 Project

ReDepress v2 : Depression Relapse Forecasting using Time
series language modelling

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Problem Statement

Objective: Predict the risk and timing of relapse for individuals with a history of mental health conditions (e.g., depression) based on their social media activity.

Input:

- A chronological sequence of user-generated social media posts (e.g., Reddit posts).
- Metadata associated with each post (optional: timestamp, subreddit, engagement metrics).

Output:

- Predicted “state of mind” at any future time.
- Forecasted time of potential relapse, defined as when the predicted state of mind crosses a critical threshold.

Baseline

ReDepress: A Cognitive Framework for Detecting Depression Relapse from Social Media

<https://arxiv.org/pdf/2509.17991>

Motivation

- Mental health relapse prediction is crucial for early intervention and personalized care.
- Traditional clinical monitoring is often limited by sparse interactions; social media provides a rich, continuous source of behavioral data.
- Automated relapse forecasting could enable timely support and reduce adverse outcomes.
- Existing research often focuses on classification (relapsed vs. non-relapsed), but predicting **time-to-relapse** provides actionable insights for preventive measures.

Dataset

<https://huggingface.co/datasets/aakash-agarwal/ReDepress>

Methodology

Overview:

We propose a two-stage approach combining **state-space modeling** and **time-series forecasting**.

Stage 1: Modeling User “State of Mind”

- Define each user’s state of mind at time t as a latent vector $\mathbf{s}_t$.
- Update rule:

$$\mathbf{s}_t = \text{NN}(\mathbf{s}_{t-1}, \text{embedding}(P_t))$$

where NN is a black-box neural network (e.g., RNN, Transformer) and $\text{embedding}(P_t)$ represents the post.

- Result: sequence of state-of-mind vectors $[\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_T]$.

Stage 2: Forecasting Time-to-Relapse

- Feed the state-of-mind sequence into a time-series forecasting model (e.g., state-space model, Transformer-based sequence predictor).
- Predict future states $\mathbf{s}_{T+1}, \mathbf{s}_{T+2}, \dots$
- Determine predicted relapse time as the first time $\mathbf{s}_t < \text{threshold}$.

Thank you